SDC Hackathon 2026 Grand Final, Domain 1: Future-Ready Workforce

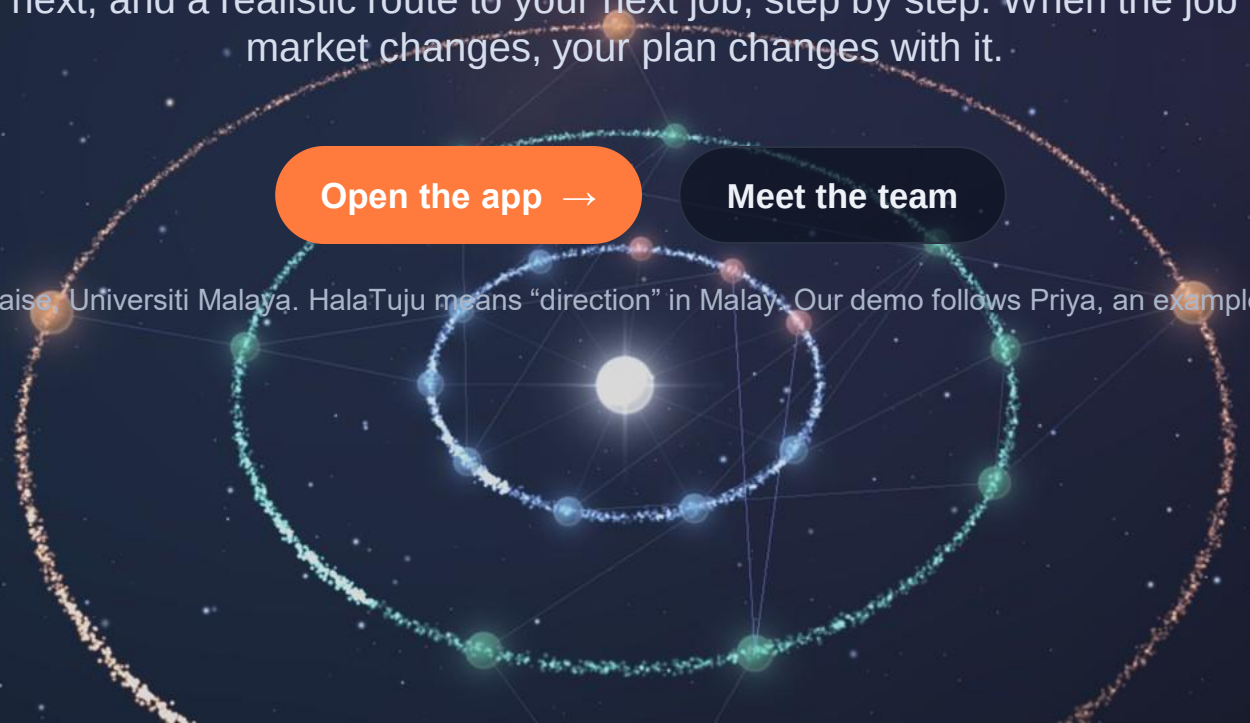
A career GPS for the AI era.

HalaTuju shows you which of your skills AI is taking over, what to learn next, and a realistic route to your next job, step by step. When the job market changes, your plan changes with it.

Open the app →

Meet the team

Team Dsaise, Universiti Malaya. HalaTuju means “direction” in Malay. Our demo follows Priya, an example worker.



Universiti Malaya, SDC Hackathon 2026 finalists

Team DSAISE

Five students from Universiti Malaya, Faculty of Computer Science and Information Technology.



Leow Shen En

Team leader



Choong Zhuo Lin

Technical lead



Tan Wei Feng

Presenter and documentation



Lim Jie Shin

Research and data



Esther Kong Yuan Er

UI/UX design

01

The problem

Will AI take my job? How much of it? Which of my skills?

Priya has been an accounts payable clerk for 11 years. Next year AI takes over invoice matching. She has six months to move, and nobody can tell her which of her skills still count.

620,000

Malaysian jobs hit hard by AI within 3–5 years.

TalentCorp, 2024

1 in 4

workers worldwide do a job AI can partly do.

ILO, 2025

56%

use AI at work, but only 26% are trained for it.

Hays, 2025

Today’s tools, such as MyMahir, MYFutureJobs and LinkedIn Career Explorer, compare your skills or suggest courses. None of them gives you a dated, step-by-step plan that updates itself when AI changes your work.

02

What we built

Three engines most career tools don’t have.

1 OUR OWN MODEL

We trained our own AI-exposure model.

Shows which of your skills AI can already do.

13,939 skills scored, trained on RunPod

✕ Most tools: a job match, no view of your own skills

2 FUTURE SIMULATION

We simulate 1,000 futures per route.

Shows how often your route holds, and what breaks it.

1,000 futures per route, plain code, no guessing

✕ Most tools: one fixed plan

3 AI RESEARCH AGENTS

AI agents keep your route up to date.

Three agents vote on each news finding. You decide.

74 trusted sites checked daily

✕ Most tools: fixed data, updated by hand

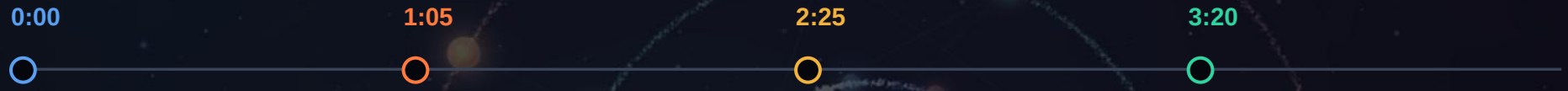
For workers, graduates, career switchers, and the HR teams that guide them.

03

Live demo

Watch Priya’s route change, live.

Four minutes in the real app: which of her skills AI takes, where she can move, how her plan is tested, and what happens when the news changes.



Stay

Her CV goes in, her personal details come out. Which of her 10 skills AI puts at risk.

Move

Her best job matches, a UN goal put first, and the route tested in 1,000 futures.

Updated

The news changes, three agents vote, and she chooses: keep my route, or switch.

For everyone

The same app in Bahasa Melayu, and the whole journey from the keyboard.

Open the app →

halatuju.replit.app

Backup video

04

How well it works

Measured, not claimed.

We trained our own AI model and tested it on jobs it had never seen, then tested how well the app reads CVs. Where we fixed something, we show the first result too.

OUR AI MODEL AGAINST HUMAN EXPERTS

0.92

agreement on 185 jobs it had never seen

1.0 means perfect agreement



Scale starts at 0.80. Built on the EU's list of 3,039 jobs and 13,939 skills. Against the ILO's separate index of 2,956 jobs, never trained on: 0.79.

READING CVS AND QUALITY

35 of 36

Current job read correctly
36 test CVs, 24 Oct 2026

154 of 155

Personal details removed first
before any AI reads the CV; 128 of 155 on the first test

95–100%

Right job in the top 3, no CV
from a few questions; finance, sales, IT

543

Automated tests
run on every code update

05

How plans stay current

How your plan stays up to date when AI changes jobs.

Agents read and vote. Fixed formulas, not AI, set every number, so each change can be undone.

1. **Read** — 74 trusted sites, daily
2. **Check** — quote must be on the page
3. **Vote** — three agents, two vendors
4. **Decide** — a fixed rule, not a model
5. **You choose** — keep my route, or switch

STEP 3 · VOTE — THREE AGENTS, TWO VENDORS

EXAMPLE “Shared-service centres move invoice matching to AI.”

Evidence checker GPT-5.5	Is it real, recent and trusted?	YES
Skill matcher GPT-6-sol	Is it really about this skill?	YES
Sceptic DeepSeek-V4-Pro	Is it hype or a sales pitch?	NO

STEP 4 · FIXED RULE — NO MODEL DECIDES

■■□	**2 or 3 yes**	Applied, and can be undone — EXAMPLE RESULT
■□□	**1 yes**	A person decides
□□□	**0 yes**	Rejected, reasons recorded

06

The six must-haves

Everything the Domain 1 brief asked for

All six must-haves run live in one journey. The right column shows where you see each one in the demo.

1	**Skills profile intake** CV, LinkedIn, scan or a few questions. Personal details removed first.	✓ Live	Act 1
2	**AI-trend signals, with source and date** 74 trusted sites. Every quote checked, three agents vote.	✓ Live	Act 3
3	**Ranked roles, with the match explained** Match score, gap in weeks, and whether demand is rising.	✓ Live	Act 2
4	**A personal pathway, with time and cost** Fastest, Balanced or Safest, fitted to her hours, with RM cost.	✓ Live	Act 2
5	**SDG mapping that changes something** Put a UN goal first and the job list re-ranks.	✓ Live	Act 2
6	**More than one future, as estimates** Slow or fast AI, 2026 to 2030, 1,000 futures per route.	✓ Live	Acts 2, 3

07

UN goals

A UN goal that changes the ranking, not just a label



Decent work

Target 8.5

Move workers whose jobs are exposed to AI into new roles, instead of laying them off.



Productivity through innovation

Target 8.2

Workers learn the AI skills their own job now needs.



Skills for work

Target 4.4

Every route is a skills plan, with hours and courses.



Any goal, by choice

Put a UN goal first

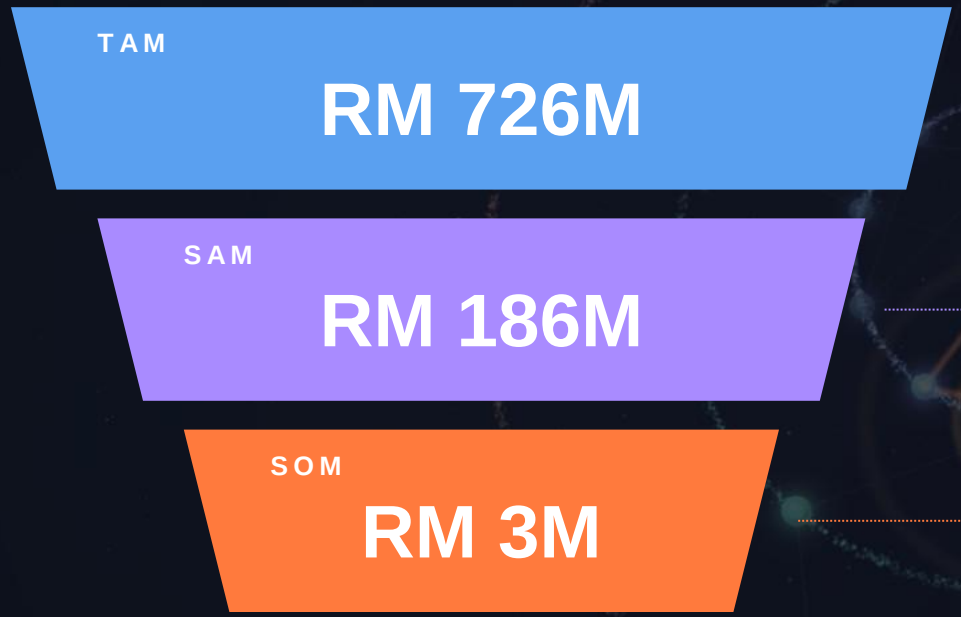
Put the goal you care about first, and the job list re-ranks around it, showing the cost in match points.

08

Impact and feasibility

A market of 620,000 workers who need this now.

Sized from the number of Malaysian workers in jobs that AI will change, at RM 300 per worker a year.



Everyone who could use it
2.42 million Malaysian workers in jobs with high or medium AI impact

Who needs it most
620,000 workers in highly impacted jobs

Our target by year three
10,000 seats at RM 300 a year. Break-even at about 1,000 seats

Sources: TalentCorp (2024), GBS Malaysia Mid-Term Report (2025), Department of Statistics Malaysia (DOSM). Prices are pilot assumptions.

09

Business model

Free for workers. Employers pay, because layoffs cost more.

FOR WORKERS

Personal

Free

- ✓ Career map and routes
- ✓ 1,000-futures test
- ✓ Private to the worker

Builds trust and reach

FOR WORKERS

Personal Plus

RM 9.90

a month

- ✓ Everything in Personal
- ✓ Certificates
- ✓ Advisor sessions

Small extra income

MAIN REVENUE

FOR EMPLOYERS

Enterprise

RM 25

per worker a month

- ✓ HR view of skills at risk
- ✓ Internal moves before layoffs
- ✓ Help for leavers
- ✓ Staff opt in, data stays theirs

One layoff avoided (RM 22,800) pays for 76 workers for a year

FOR GOVERNMENT

Public partners

Licence

a year, per agency or campus

- ✓ PERKESO
- ✓ TalentCorp
- ✓ Universities

Reach at national scale

Severance: 20 days' wages per year of service at the DOSM 2025 clerical median, RM 2,693. Reading a CV costs us RM 0.03 and a news check RM 1.05, measured. Personal data handled under PDPA. Prices are pilot assumptions.

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Roadmap

KL first. Break-even in 2027. RM 1.9M a year by 2029.

Now, Oct 2026

Live and tested

Any of 3,039 jobs, 1,000-futures test, three-agent vote, 543 tests.

Done

Next 3 months

Ready for employers

Track all 13,939 skills daily. Import HR data. HRD Corp-claimable courses.

Goal: letter of intent from 1 employer

H1 2027

Pilot in Greater KL

Shared-service centres around KL Sentral, Bangsar South and Cyberjaya. First talks with MDEC and GBS Malaysia.

Goal: 2–3 centres, about 1,500 staff

Oct 2027

Break-even

Monthly profit turns positive in the median of 10,000 simulated futures.

Funding needed: about RM 150k

2028–29

Scale

More employers and universities in the Klang Valley, then Penang and Johor. Licensed Malaysian job-ad data. More sectors.

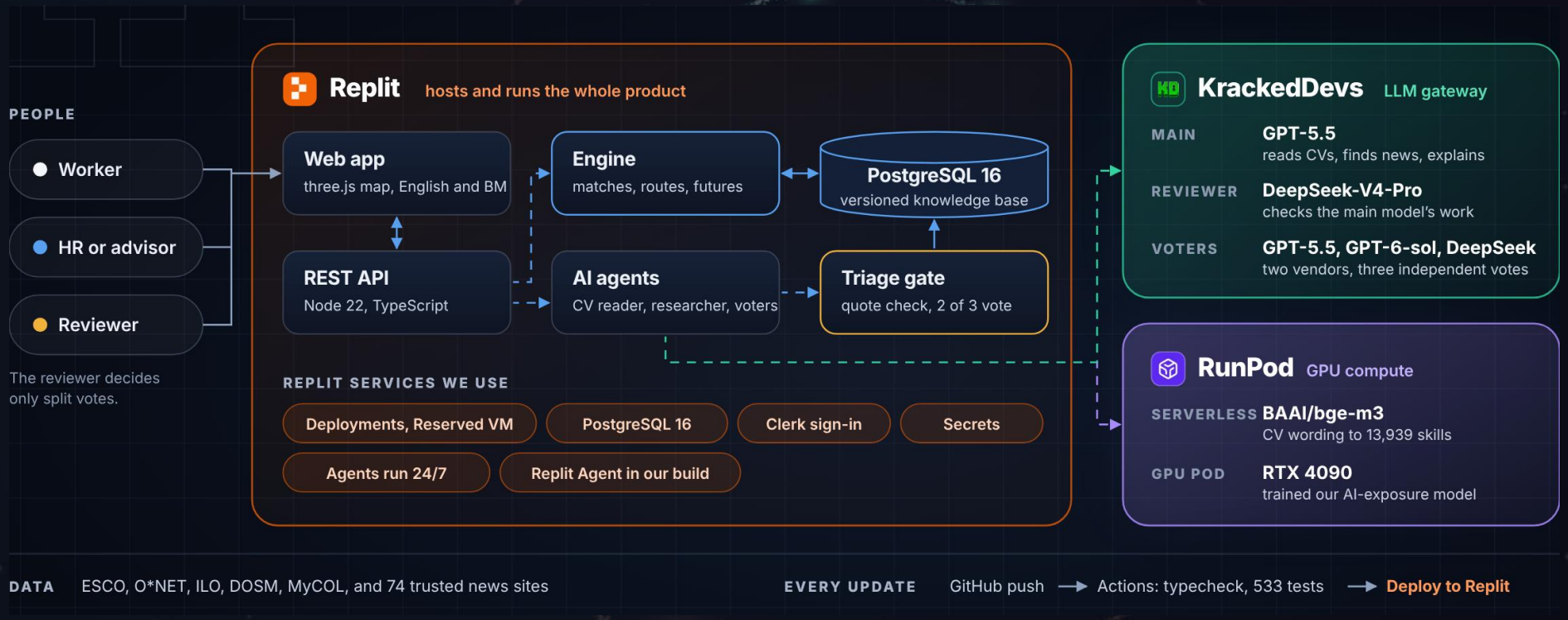
Median: 7,400 seats, RM 1.9M a year

OUR ASK Introductions to one pilot employer, and to HRD Corp so our courses can be claimed.

11

Architecture

One app on Replit, with KrackedDevs and RunPod plugged in.



HalaTuju, Team Dsaise, Universiti Malaya

Knowing what is coming is one thing.

Knowing what to do next is another.

Thank you.

Open the app →

Source code



Scan to try it
halatuju.replit.app

Leow Shen En, Choong Zhuo Lin, Tan Wei Feng, Lim Jie Shin, Esther Kong Yuan Er

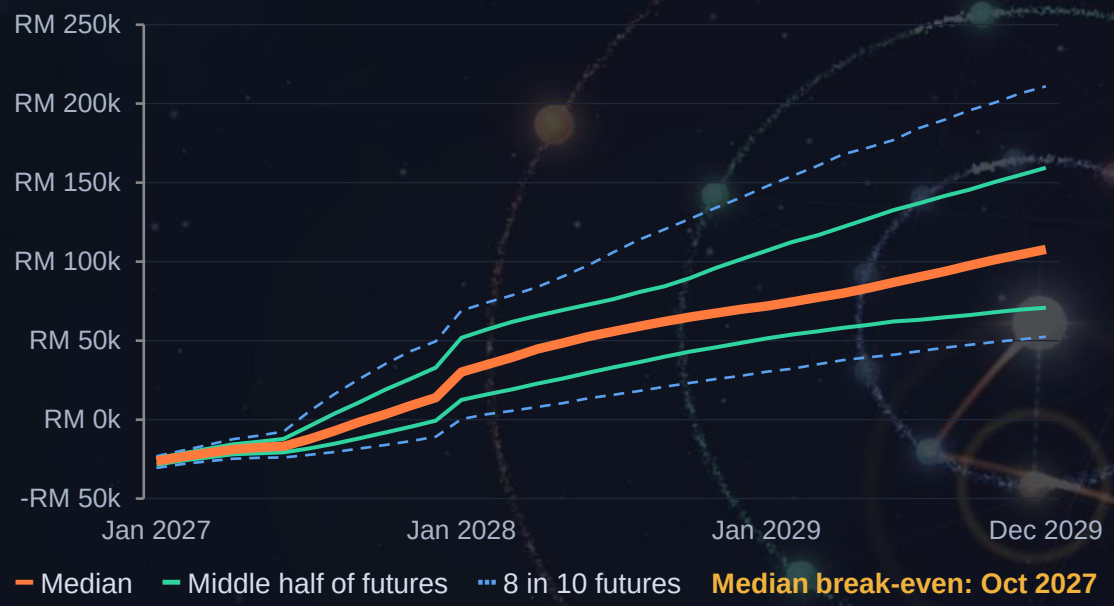
A1

Appendix: business stress test

We ran the business 10,000 times.

The same stress test we give workers, applied to our own company. Every input is a range, not a single guess.

MONTHLY PROFIT, 2027 TO 2029 (RM)



98%
of futures break even within 18 months

RM 152k
funding needed to get there (1 in 10 futures: RM 244k)

RM 1.9M
yearly revenue by end of 2029 (80% range RM 1.0M–3.2M)

7,358
seats by end of 2029; our 10,000 target is a stretch, reached in 1 of 4 futures

WHAT MOVES YEAR-THREE PROFIT MOST

Staff per employer	Employers signed a month	Price we actually charge	Employers who leave
RM 2.1M swing	RM 1.6M swing	RM 552k swing	RM 460k swing

Inputs: RM 25 a seat a month, RM 0.50 to serve, RM 24,452 fixed a month; 1 to 4 pilot employers at 30–80% of price; 0.25–0.9 new employers a month of 200–900 staff; 0.5–3% monthly churn; team grows to 5 past 5,000 seats. Seeded, 10,000 runs.

A2

Appendix: how well it works

Three tests that could have embarrassed us.

We ran them anyway, and we show the results as they came out.

01

Fairness test

We gave the same CV six different names, genders, ages, races, religions and marital statuses. The AI received exactly the same text every time, so the skills, matches and routes came out the same.

tests/fairness.test.ts

02

Tamper test

Text hidden in a web page cannot change what the AI reads from a CV, invent data, or skip our checks. Page text is treated as data, never as instructions.

tests/injection.test.ts

03

Live run, 23 Oct 2026

Our agents read 769 articles in 5.5 minutes and sent 7 findings to a person. Every change is saved as a version and can be undone.

Measured on the live server

A3

Appendix: judges, try to break it

Change the data. Check a course. Ask why.

Three things you can test in the app right now, and what you will see.

01

Change the trend data now

Open the agents panel and watch three agents vote on a news finding. The knowledge base moves from version 1 to version 2, every plan is re-scored, and the route visibly changes on the map, with the source attached. Undo puts it back.

In rehearsal, the route changed in 0.1 seconds

02

Is this course real?

30 real course listings with price and hours. Jobs come from the EU’s official list; wages are Malaysian government figures from DOSM. When no course exists, the app says “no course listed yet” instead of making one up.

Click any step to open the listing

03

Why this score, and what would raise it?

The breakdown lists each skill the job needs, how much employers want it, and your level. Confirm a skill or finish a step, and the number updates in front of you.

A fixed formula. No AI in the loop.

A4

Appendix: what we say openly

What this prototype cannot do yet.

Sample data in places

The 16-job catalogue in the demo uses labelled sample data. “Check any job” uses the real official list. Most study times are estimates, and our task evidence comes from US data.

Matching is still rough

A nurse matches her own job at only about 30%, because the official list has about 60 skills per job and a CV names about 16. The questions reach about 70% accuracy for trade jobs.

Tests still to run

We have not yet tested clues such as school name or graduation year. The 1,000 futures test a plan; they do not predict any one person’s future.